

# HPE Networking Assistant — Test Plan

End-to-end acceptance test plan covering the **installation process** and **every tool and visualizer**. Written to be executed on a single Windows PC acting as the test client.

- **Product version under test:** 1.9.0
- **Repository:** <https://github.com/hpe-networking-lab/hpe-networking-assistant>
- **Two test layers:**
  1. **Automated** — the pytest suite (65 tests) validates all tool logic with a mocked Mist API; no token or network required.
  2. **Manual acceptance** — live tests against a real Mist tenant through Claude Desktop, below.
  3. **Companion dashboards** — two live Cowork artifacts (Section 9a), separate from the packaged .dxt.

Mark each case **PASS / FAIL** in the result column and capture notes/screenshots for failures.

## 1. Test environment

| Item                          | Value for this run                                    |
|-------------------------------|-------------------------------------------------------|
| Test client                   | This Windows PC                                       |
| OS                            | Windows 10/11                                         |
| Python                        | 3.10+ (python --version)                              |
| Claude Desktop                | Latest                                                |
| Mist tenant                   | _Use a <b>lab/non-production</b> org for write tests_ |
| Mist token role (read tests)  | Observer / read                                       |
| Mist token role (write tests) | Network Admin / write                                 |
| Region                        | Auto-detected                                         |

### Safety

- **Read-only tools are safe in production.** Run them anywhere.
- **Write-mode tests (Section 9) change device state/config** (rename, reboot, LED, claim/assign). Run them **only against a lab org** and only on devices you may disrupt.

## 2. Prerequisites checklist

| #  | Check                                               | Expected            | Result |
|----|-----------------------------------------------------|---------------------|--------|
| P1 | python --version                                    | 3.10 or newer       |        |
| P2 | Claude Desktop installed and opens                  | Launches            |        |
| P3 | Mist API token created (My Account → API Token)     | Token copied        |        |
| P4 | (Optional) Git installed for the dev/automated path | git --version works |        |

### 3. Automated test suite (developer path)

Run from a clone of the repo. Validates all tools/visualizers offline.

| #  | Step              | Command                                                               | Expected                | Result |
|----|-------------------|-----------------------------------------------------------------------|-------------------------|--------|
| A1 | Clone + install   | <code>pip install -e ".[dev]"</code>                                  | Installs with no errors |        |
| A2 | Run tests         | <code>pytest -q</code>                                                | <b>90 passed</b>        |        |
| A3 | Byte-compile      | <code>python -m compileall src server</code>                          | No errors               |        |
| A4 | Validate manifest | <code>python -c "import json;json.load(open('manifest.json'))"</code> | No error                |        |

Coverage map (automated):

| Area                                           | Test file                                                                                                                                      |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Region normalization + detection               | <code>test_regions.py</code> , <code>test_discovery.py</code>                                                                                  |
| Mist client (auth, pagination, writes, search) | <code>test_mist_client.py</code> , <code>test_write_mode.py</code> , <code>test_client_trace.py</code> , <code>test_access_assurance.py</code> |
| MCP server / dispatch / tool registry          | <code>test_server.py</code>                                                                                                                    |
| Onboarding & status                            | <code>test_onboarding.py</code> , <code>test_status.py</code>                                                                                  |
| Validation framework                           | <code>test_validation.py</code>                                                                                                                |
| Reports                                        | <code>test_reports.py</code>                                                                                                                   |
| NAC visualizer                                 | <code>test_nac_visualizer.py</code>                                                                                                            |

### 4. Installation tests (TC-INST)

| #      | Objective                  | Steps                                                                                   | Expected                                                                                                        | Result |
|--------|----------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------|
| INST-1 | Download artifact          | From the latest release, download <code>hpe-networking-assistant-&lt;ver&gt;.dxt</code> | File downloads                                                                                                  |        |
| INST-2 | Verify checksum (optional) | Compare SHA-256 against <code>SHA256SUMS.txt</code>                                     | Matches                                                                                                         |        |
| INST-3 | Install extension          | Claude Desktop → Settings → Extensions → drag in the <code>.dxt</code>                  | Extension details + config form appear                                                                          |        |
| INST-4 | Config form is token-only  | Inspect the form                                                                        | Only <b>Mist API Token</b> (required) and <b>Enable write operations</b> (off) are shown — no region/org fields |        |
| INST-5 | Enter token + enable       | Paste token, leave write off, Install/Enable                                            | Extension enables without error                                                                                 |        |

| #      | Objective             | Steps                                           | Expected                                                   | Result |
|--------|-----------------------|-------------------------------------------------|------------------------------------------------------------|--------|
| INST-6 | Token stored securely | —                                               | Token saved to OS keychain (not shown again in plain text) |        |
| INST-7 | Time budget           | From clicking install to first successful query | <b>Under 10 minutes</b>                                    |        |

## 5. Onboarding tests (TC-ONB)

| #     | Objective            | Prompt / action                                 | Expected                                                                                                                         | Result |
|-------|----------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--------|
| ONB-1 | First-run wizard     | New chat: <b>"Set me up."</b>                   | Claude calls <code>start_setup</code> ; auto-detects region, discovers orgs/sites, runs validation, returns <b>READY FOR USE</b> |        |
| ONB-2 | Region auto-detected | Observe report                                  | Correct region shown; you never entered it                                                                                       |        |
| ONB-3 | No IDs required      | Observe                                         | You are never asked for Org ID / Site ID / API endpoint                                                                          |        |
| ONB-4 | Multi-org prompt     | (If token sees >1 org)                          | Claude lists orgs <b>by name</b> and asks which to use; answering by name completes setup                                        |        |
| ONB-5 | Persistence          | Restart Claude Desktop, ask "what's my status?" | Region/org remembered (no re-probe needed)                                                                                       |        |

## 6. Read-only tool tests (TC-RO)

Use natural-language prompts; verify Claude calls the right tool and returns sensible data.

| #     | Tool                                   | Prompt                                                      | Expected                                              | Result |
|-------|----------------------------------------|-------------------------------------------------------------|-------------------------------------------------------|--------|
| RO-1  | <code>get_status</code>                | "Am I in read-only or read-write mode?"                     | Reports read-only, region, org, account               |        |
| RO-2  | <code>get_organizations</code>         | "What organizations can I access?"                          | Lists org name(s)                                     |        |
| RO-3  | <code>get_sites</code>                 | "List my sites."                                            | Site names + locations                                |        |
| RO-4  | <code>get_access_points</code>         | "Show all APs in my organization." (success-criteria query) | AP list with online/offline counts                    |        |
| RO-5  | <code>get_access_points</code> (site)  | "Show APs at <site name>."                                  | APs filtered to that site                             |        |
| RO-6  | <code>get_switches</code>              | "List my switches."                                         | Switch list with status                               |        |
| RO-7  | <code>get_clients</code>               | "How many clients are connected at <site>?"                 | Client count/list                                     |        |
| RO-8  | <code>get_offline_access_points</code> | "Which access points are offline right now?"                | Offline AP list (matches portal)                      |        |
| RO-9  | Cross-check                            | Compare RO-4/RO-8 counts to the Mist portal                 | Numbers match                                         |        |
| RO-10 | <code>get_wired_clients</code>         | "List wired clients / what's on switch port X?"             | Wired clients with switch MAC, port, VLAN, IP, vendor |        |

| #     | Tool               | Prompt                                           | Expected                                                                                        | Result |
|-------|--------------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------|--------|
| RO-11 | get_marvis_actions | "What's wrong with my network?"                  | Marvis suggested actions + fixes, per-category counts; status=open filters to active            |        |
| RO-12 | get_alarms         | "Show critical alarms in the last day."          | Alarms with per-severity/per-type counts                                                        |        |
| RO-13 | get_sle            | "How's the user experience by site?"             | Per-site SLE metrics as percentages                                                             |        |
| RO-14 | get_switch_ports   | "Show ports on switch <mac> / down ports"        | Port link/speed/PoE/neighbor; filters by switch/site/up                                         |        |
| RO-15 | export_org_config  | "Back up my org config to a file."               | JSON bundle (org + sites/networks/templates/WLANs/NAC/...) with per-type counts; saved as .json |        |
| RO-16 | set_active_org     | "Use the <name> organization." (multi-org token) | Active org set by name; later tools target it                                                   |        |
| RO-17 | diff_org_config    | "Compare current config to last week's backup."  | Added/removed/changed per resource type; "no drift" when identical                              |        |

## 7. Report tests (TC-RPT)

| #     | Tool                      | Prompt                                                    | Expected                                                                  | Result |
|-------|---------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------|--------|
| RPT-1 | generate_health_report    | "Generate a network health report and save it as a file." | Markdown report (totals, offline APs, per-site) saved as .md              |        |
| RPT-2 | Health accuracy           | Open the report                                           | Counts match RO-4/RO-6/RO-8                                               |        |
| RPT-3 | Skip clients              | "Generate a health report without the client count."      | Report notes clients "not collected" (faster)                             |        |
| RPT-4 | generate_inventory_report | "Create an inventory report of all devices."              | Markdown table: every AP+switch with model/serial/MAC/site/status/version |        |
| RPT-5 | Convert                   | "Convert that report to PDF."                             | Claude produces a PDF from the Markdown                                   |        |
| RPT-6 | generate_firmware_report  | "Which devices are behind on firmware?"                   | Per-model target version + list of devices behind it                      |        |

## 8. Client trace & Access Assurance tests (TC-CT / TC-AA)

> Pick a real client MAC/hostname from RO-7 for these.

| #    | Tool                   | Prompt                                   | Expected                                              | Result |
|------|------------------------|------------------------------------------|-------------------------------------------------------|--------|
| CT-1 | find_client (mac)      | "Find the client with MAC <mac>."        | Site, AP, SSID, IP, band, RSSI, last seen             |        |
| CT-2 | find_client (hostname) | "Where is hostname <name>?"              | Matching client(s) located                            |        |
| CT-3 | trace_client           | "Trace client <mac> over the last hour." | Event timeline, per-type counts, highlighted failures |        |
| CT-4 | No-input guard         | "Find a client." (no mac/hostname)       | Friendly error asking for mac or hostname             |        |

| #    | Tool                        | Prompt                                                       | Expected                                                                                                                 | Result |
|------|-----------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------|
| AA-1 | get_nac_clients             | "List NAC clients from the last day."                        | Authenticated NAC clients (user, type, auth type, SSID, VLAN, rule, status) — _empty if Access Assurance not configured_ |        |
| AA-2 | Filter                      | "Show only EAP-TLS NAC clients."                             | Filtered list                                                                                                            |        |
| AA-3 | troubleshoot_authentication | "Troubleshoot authentication failures in the last 24 hours." | Event timeline, type counts, failure highlights                                                                          |        |
| AA-4 | Focus by client (MAC)       | "Why is <mac> failing authentication?"                       | Events scoped to that MAC                                                                                                |        |
| AA-5 | Focus by user (user)        | "Why is bob@corp failing 802.1X?"                            | Events scoped to that username/cert CN (free-text), failures highlighted                                                 |        |

> **Note:** AA tests require a tenant with **Access Assurance (NAC)** enabled. If your tenant has no NAC, expect empty results (not an error). Confirm the NAC events endpoint path against your tenant (see Release Notes 1.6.0).

## 9. NAC dashboard test (TC-NAC)

| #     | Tool                   | Prompt                                       | Expected                                                                                                                            | Result |
|-------|------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--------|
| NAC-1 | generate_nac_dashboard | "Build a NAC dashboard and save it as HTML." | Self-contained .html produced                                                                                                       |        |
| NAC-2 | Open offline           | Double-click the .html                       | Opens in browser; cards + bar charts render with <b>no internet</b>                                                                 |        |
| NAC-3 | Content                | Inspect                                      | Cards (clients, events, failures, success rate) + charts for auth types, client types, status, event types, top failing users/rules |        |
| NAC-4 | Empty-state            | (No-NAC tenant)                              | Renders with zeros / "No data" rather than erroring                                                                                 |        |

### 9a. Companion Cowork dashboards (TC-CD)

> Live, connector-backed dashboards in Claude Cowork, separate from the packaged extension. They require the Mist connector to be connected in Cowork.

| #    | Objective                  | Action                                              | Expected                                                                          | Result |
|------|----------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------|--------|
| CD-1 | Open network/NAC dashboard | Open the "Mist Network & Access Assurance" artifact | Device-status doughnut + cards load from live data                                |        |
| CD-2 | Auto-refresh               | Leave it open ~1 min                                | Updates on its own (~30s cadence); "last updated" time advances; pulsing live dot |        |
| CD-3 | NAC empty-state            | (No-NAC tenant)                                     | Shows "Access Assurance may not be configured" rather than erroring               |        |
| CD-4 | Open NAC Auth Debugger     | Open the "NAC Auth Debugger" artifact               | Loads with input + time-window controls                                           |        |

| #    | Objective         | Action                                          | Expected                                                              | Result |
|------|-------------------|-------------------------------------------------|-----------------------------------------------------------------------|--------|
| CD-5 | Debug by username | Type a username, choose a window, click Debug   | Pulls that identity's NAC events; pass/fail counts; timeline          |        |
| CD-6 | Debug by MAC      | Enter a 12-hex MAC                              | Auto-routes to MAC filter; events returned                            |        |
| CD-7 | Failure decode    | (Identity with a failure)                       | "Most recent failure" card shows reason, matched rule, auth type, NAS |        |
| CD-8 | Explain hand-off  | Click "Ask Claude to explain and suggest a fix" | Failure detail is sent to chat as a prompt                            |        |

## 10. Write-mode tests (TC-WR) — LAB ORG ONLY

> These change device state/config. Use a lab org and a **write-capable token**.

| #    | Objective          | Prompt / action                                                     | Expected                                                                                         | Result |
|------|--------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------|
| WR-1 | Enable write mode  | Settings → toggle <b>Enable write operations</b> on; restart prompt | Extension restarts                                                                               |        |
| WR-2 | Write tools appear | "What can you do now?" / list tools                                 | rename_device, reboot_device, locate_device, claim_devices, assign_devices_to_site now available |        |
| WR-3 | Write-readiness    | "Check my setup." (get_status)                                      | If token is read-only → reports it + guidance; if write-capable → confirms write access          |        |
| WR-4 | Confirmation gate  | "Rename AP <mac> to TEST-AP."                                       | Claude asks to confirm; nothing changes until you confirm                                        |        |
| WR-5 | Rename             | Confirm WR-4                                                        | Device renamed (verify in portal); revert afterward                                              |        |
| WR-6 | LED locate         | "BI                                                                 |                                                                                                  |        |